

Exercise 4

SQL Date Functions and Date math

• Questions 1-5 - Extract Date parts

• extract individual pieces from a date column - the day name, month name, Month number, year and day of the month

① Day name from order date

Table: orders

order_id	customer_id	order_date
1001	101	2026-05-01
1002	102	2026-05-02
1003	103	2026-05-03
1004	104	2026-05-04
1005	105	2026-05-05

• Return each order with the day name of the order date.

SQL: `SELECT order_id, customer_id, order_date,
DAYNAME (order_date) AS day_name
FROM orders;`

Output	order_id	customer_id	order_date	day_name
	1001	101	2026-05-01	Friday
	1002	102	2026-05-02	Saturday
	1003	103	2026-05-03	Sunday
	1004	104	2026-05-04	Monday
	1005	105	2026-05-05	Tuesday

1.2 Month Name This Sign up date

Table Customer_Signup

Customer-id	Customer-Name	Signup-date
201	John	2026-01-01
202	Amy	2026-02-01
203	Peter	2026-03-01
204	Sarah	2026-04-01
205	Thomas	2026-05-01

• Return each customer with the name of the month in which they signed up

Query: **Select** Customer-id, Customer-name, Signup-date
 Monthname(Signup-date) As Signup-month-name
From Customer_Signup

Output	Customer-id	Customer-name	Signup-date	Signup-month-name
	201	John	2026-01-01	January
	202	Amy	2026-02-01	February
	203	Peter	2026-03-01	March
	204	Sarah	2026-04-01	April
	205	Thomas	2026-05-01	May

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1-2 months name from sign up date

Table Customer_Signups

Customer-id	customer-name	Signup-date
201	John	2026-01-15
202	Mary	2026-02-20
203	Peter	2026-03-05
204	Sarah	2026-04-18
205	Thabo	2026-05-30

Return each customer with the name of the month in which they signed up.

Syntax:
 SELECT customer-id, customer-name, Signup-date,
 MONTHNAME (Signup-date) AS Signup-month-name
 FROM Customer_Signups;

Output	Customer-id	Customer-name	Signup-date	Signup-month-name
	201	John	2026-01-15	January
	202	Mary	2026-02-20	February
	203	Peter	2026-03-05	March
	204	Sarah	2026-04-18	April
	205	Thabo	2026-05-30	May

2026-01-10	laptop	12000
2026-02-15	mouse	250
2026-03-20	keyboard	700
2026-04-25	monitor	35000
2026-05-30	desk	2000

Table: Customer - Signups

12 Month Name From Sign up date

(3) Month number From Sale date

Table: Sales

Sale_id	Product_name	Sale_date	amount
S001	laptop	2026-01-10	12000
S002	mouse	2026-02-15	250
S003	keyboard	2026-03-20	700
S004	monitor	2026-04-25	35000
S005	Desk	2026-05-30	2000

Return each Sale with the month number extracted From Sale_date

Syntax:
 SELECT Sale_id, Product_name, Sale_date,
 MONTH (Sale_date) AS Sale_month
 FROM Sales;

Output	Sale_id	Product_name	Sale_date	Sale_month
	S001	laptop	2026-01-10	1
	S002	mouse	2026-02-15	2
	S003	keyboard	2026-03-20	3
	S004	monitor	2026-04-25	4
	S005	Desk	2026-05-30	5

1.1.1 Year From transaction date

Table : Transactions

Transaction_id	Customer_id	transaction_date	amount
T001	101	2024-12-15	500
T002	102	2025-01-20	1200
T003	103	2025-06-10	800
T004	104	2026-02-05	1500
T005	105	2026-05-25	2000

- Return each transaction with the year extracted From transaction_date

Syntax `SELECT transaction_id, customer_id, transaction_date,
YEAR (transaction_date) AS transaction_year,
FROM transactions;`

Output	Transaction_id	Customer_id	transaction_date	transaction_year
	T001	101	2024-12-15	2024
	T002	102	2025-01-20	2025
	T003	103	2025-06-10	2025
	T004	104	2026-02-05	2026
	T005	105	2026-05-25	2026

15) Day of month from delivery date

Table: deliveries

delivery_id	customer_id	delivery_date
D001	101	2026-05-01
D002	102	2026-05-08
D003	103	2026-05-15
D004	104	2026-05-22
D005	105	2026-05-29

- Return each delivery with the day of the month extracted from delivery date

Syntax

```
SELECT delivery_id, customer_id, delivery_date
      DAY (delivery_date) AS day-of-month
FROM deliveries
```

Output

delivery_id	customer_id	delivery_date	day-of-month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

- Questions 6-10 - Current date, Convert & Format

6) Add Today's date

Table: employees

employee_id	employee_name	department
301	Nandi	Sales
302	Brian	IT
303	Lerato	Finance
304	Sipho	HR
305	Aisha	Marketing

- Return all employees and add a column showing today's date

Syntax `SELECT employee_id, employee_name, department,
CURRENT_DATE() AS today_date
FROM employees;`

Output	employee_id	employee_name	department	today_date
	301	Nandi	Sales	2026-07-19
	302	Brian	IT	2026-07-19
	303	Lerato	Finance	2026-07-19
	304	Sipho	HR	2026-07-19
	305	Aisha	Marketing	2026-07-19

Questions 6-10 - Current date, Current & format

to Add today's date

(2) Convert text to date

Table: online-orders

Order_id	Customer_id	Order_date_text
4001	101	2026-01-15
4002	102	2026-02-20
4003	103	2026-03-25
4004	104	2026-04-10
4005	105	2026-05-05

- Convert the order_date_text String column into proper date column

Syntax `SELECT order_id, Customer_id, order_date_text,
TO_DATE (order_date_text, 'YYYY-MM-DD') AS order_date
FROM online_orders;`

Output	Order_id	Customer_id	order_date_text	order_date
	4001	101	2026-01-15	2026-01-15
	4002	102	2026-02-20	2026-02-20
	4003	103	2026-03-25	2026-03-25
	4004	104	2026-04-10	2026-04-10
	4005	105	2026-05-05	2026-05-05

4) Format date as Text

Table payment_dates

Payment_id	Customer_id	Payment_date
P001	101	2026-01-05
P002	102	2026-02-10
P003	103	2026-03-15
P004	104	2026-04-20
P005	105	2026-05-25

- Format payment_date as a text using String in the Format
YYYY-MM-DD

SQL: SELECT payment_id, customer_id, payment_date,
TO_CHAR(payment_date, 'YYYY-MM-DD') AS formatted_payment_date
FROM payment_dates;

Output	Payment_id	Customer_id	Payment_date	Formatted_payment_date
	P001	101	2026-01-05	2026-01-05
	P002	102	2026-02-10	2026-02-10
	P003	103	2026-03-15	2026-03-15
	P004	104	2026-04-20	2026-04-20
	P005	105	2026-05-25	2026-05-25

19 days since last purchase

Table: customer_purchase

customer_id	customer_name	last_purchase_date
S01	John	2026-05-01
S02	Mary	2026-05-10
S03	Peter	2026-05-15
S04	Sarah	2026-05-20
S05	Thabo	2026-05-25

- Calculate how many days have passed since each customer's last purchase. use today's date

Sqlite `SELECT customer_id, customer_name, last_purchase_date,
DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase
FROM customer_purchase;`

Output	customer_id	customer_name	last_purchase_date	days_since_last_purchase
	S01	John	2026-05-01	79
	S02	Mary	2026-05-10	70
	S03	Peter	2026-05-15	65
	S04	Sarah	2026-05-20	60
	S04	Thabo	2026-05-25	55

1.10 Expected delivery date

Table: Shipping-orders

Order_id	Customer_id	order_date
6001	101	2026-05-01
6002	102	2026-05-03
6003	103	2026-05-05
6004	104	2026-05-07
6005	105	2026-05-09

- Calculate the expected delivery date by adding 7 days to the order date.

Sqlrite
 SELECT order_id, customer_id, order_date,
 DATEADD(DAY, 7, order_date) AS expected_delivery_date
 FROM Shipping-orders;

Output	Order_id	Customer_id	order_date	expected_delivery_date
	6001	101	2026-05-01	2026-05-08
	6002	102	2026-05-03	2026-05-10
	6003	103	2026-05-05	2026-05-12
	6004	104	2026-05-07	2026-05-14
	6005	105	2026-05-09	2026-05-16

Question 11-15, Multi-Extract & date Filtering

11) Year, month, day Separately

Table: Bookings

booking_id	customer_id	booking_date
B001	101	2026-01-12
B002	102	2026-02-18
B003	103	2026-03-22
B004	104	2026-04-09
B005	105	2026-05-27

Extract the year, month number, and day from booking_date into 3 three separate columns

Syntax `SELECT booking_id, customer_id, booking_date,
 YEAR(booking_date) AS booking_year,
 MONTH(booking_date) AS booking_month,
 DAY(booking_date) AS booking_day
 FROM bookings`

Output	booking_id	customer_id	booking_date	booking_year	booking_month	booking_day
	B001	101	2026-01-12	2026	1	12
	B002	102	2026-02-18	2026	2	18
	B003	103	2026-03-22	2026	3	22
	B004	104	2026-04-09	2026	4	9
	B005	105	2026-05-27	2026	5	27

101	2024-12-15	500	2024
102	2025-01-20	1200	2025
103	2025-06-10	800	2025
104	2026-02-05	1500	2026
105	2026-05-25	2000	2026

Table: yearly_orders

Filter orders from 2026

Filter Orders From 2026

Table: yearly_orders

Order_id	Customer_id	Order_date	Amount
7001	101	2024-12-15	500
7002	102	2025-01-20	1200
7003	103	2025-06-10	800
7004	104	2026-02-05	1500
7005	105	2026-05-25	2000

Return only orders From the year 2026

Syntax
 SELECT Order_id, Customer_id, Order_date,
 YEAR (Order_date) AS Order_year, amount
 FROM yearly_orders
 WHERE YEAR (Order_date) = 2026;

Output	Order_id	Customer_id	Order_date	Order_year	Amount
	7004	104	2026-02-05	2026	1500
	7005	105	2026-05-25	2026	2000

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10) Filter orders From March

Table: monthly_orders

order_id	customer_id	order_date	amount
8001	101	2026-01-15	500
8002	102	2026-02-20	1200
8003	103	2026-03-10	800
8004	104	2026-03-25	1500
8005	105	2026-05-30	2000

Return only orders placed in March

Syntax `SELECT order_id, customer_id, order_date
MONTH (order_date) AS order_month, amount
FROM monthly_orders
WHERE MONTH (order_date) = 3;`

Output

order_id	customer_id	order_date	order_month	amount
8003	103	2026-03-10	3	800
8004	104	2026-03-25	3	1500

IS first day of the month
Table: Campaign - Sends

Send_id	Customer_id	Send_date
101	101	2026-01-10
102	102	2026-02-15
103	103	2026-03-20
104	104	2026-04-25
105	105	2026-05-30

and last day of the month

Table: Subscription

Subscription_id	Customer_id	Start_date
SUB001	101	2026-01-10
SUB002	102	2026-02-15
SUB003	103	2026-03-20
SUB004	104	2026-04-25
SUB005	105	2026-05-30

• Return the last day of the month For each Subscription Start date

Syntax
SELECT Subscription_id, customer_id, start_date,
LAST_DAY(start_date) AS month_end_date
FROM Subscriptions;

Output	Subscription_id	Customer_id	Start date	Month_end_date
	SUB001	101	2026/01/10	2026-01-31
	SUB002	102	2026-02-15	2026-02-28
	SUB003	103	2026-03-20	2026-03-31
	SUB004	104	2026-04-25	2026-04-30
	SUB005	105	2026-05-30	2026-05-31

IS First day of the month

Table: Campaign_Sends

Send_id	Customer_id	Send_date
C001	101	2026-01-12
C002	102	2026-02-18
C003	103	2026-03-22
C004	104	2026-04-09
C005	105	2026-05-07

• Return the First day of the month for each Campaign Send date.

Syntax
SELECT Send_id, Customer_id, Send_date,
DATE_TRUNC('month', Send_date) AS month_start_date
FROM Campaign_Sends;

Output	Send_id	Customer_id	Send_date	month_start_date
	C001	101	2026-01-12	2026-01-01
	C002	102	2026-02-18	2026-02-01
	C003	103	2026-03-22	2026-03-01
	C004	104	2026-04-09	2026-04-01
	C005	105	2026-05-07	2026-05-01

1998-01-05	101	Customer ID
1998-02-10	102	Customer ID
1998-03-15	103	Customer ID
1998-04-20	104	Customer ID
1998-05-25	105	Customer ID

Table Customer_Bu_in_days

11 Customer age from date of birth

is format as month and year

Table Invoice_dates

Invoice_id	Customer_id	Invoice_date
INV001	101	2026-01-05
INV002	102	2026-02-10
INV003	103	2026-03-15
INV004	104	2026-04-20
INV005	105	2026-05-25

Format Invoice_date as a text string showing month name and year

Syntax `SELECT Invoice_id, Customer_id, Invoice_date, TO_CHAR(Invoice_date, 'Month YYYY') AS Invoice_month_year FROM Invoice_dates;`

Output	Invoice_id	Customer_id	Invoice_date	Invoice month year
	INV001	101	2026-01-05	January 2026
	INV002	102	2026-02-10	February 2026
	INV003	103	2026-03-15	March 2026
	INV004	104	2026-04-20	April 2026
	INV005	105	2026-05-25	May 2026

Customer age From date of birth

Table Customer_Birthdays

Customer_id	Customer-Name	date of Birth
901	John	1998-05-10
902	Mary	1990-08-20
903	Peter	2002-03-15
904	Sarah	1985-12-01
905	Thabo	2000-07-30

• Calculate each customers age in years using their date of birth and to day's date.

Syntax
 SELECT Customer-id, Customer-Name, date-of-birth,
 FLOOR (DATE DIFF (CURRENT-DATE (), date-of-birth/365.25))
 AS customer_age
 FROM Customer - Birthdays;

Out put

Customer-id	Customer-name	date-of-birth	Customer-age
901	John	1998-05-10	28
902	Mary	1990-08-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
904	Thabo	2000-07-30	25

• Classify orders as weekday

Table: weekend_orders

order id	customer id	order date
9001	101	2026-05-01
9002	102	2026-05-02
9003	103	2026-05-03
9004	104	2026-05-04
9005	105	2026-05-05

- Classify each order as weekend or weekday based on the day name of the order date. Saturday and Sunday = weekend. All others = weekdays

Syntax: SELECT

orderdate,

CASE

WHEN Dayname (orderdate) in ('Saturday', 'Sunday') THEN 'weekend'

ELSE 'weekday'

END AS Day-type

FROM orders

order-date	day-type
2026/05/01	weekday
2026/05/02	weekend
2026/05/03	weekend
2026/05/04	weekend
2026/05/05	weekday

19 Transactions by Quarter

Table: quarterly_transactions

Transaction_id	customer_id	transaction_date	amount
Q001	101	2026-01-15	500
Q002	102	2026-03-20	1200
Q003	103	2026-04-10	800
Q004	104	2026-07-05	1500
Q005	105	2026-10-25	2000

- Extract the quarter number From each transaction date.

```
SELECT transaction_id,
       customer_id,
       transaction_date,
       amount,
       quarter(transaction_date) AS quarter_number
```

```
FROM quarterly_transactions;
```

Q005	2026-04-01	101	500
Q004	2026-04-10	102	1200
Q003	2026-05-14	103	500
Q002	2026-05-10	104	1500
Q001	2026-06-28	105	2000

Table: quarterly_transactions

19 Transactions by Quarter

20 Orders older than 30 days

Table: recent orders

Order id	Customer id	Order date	Amount
R001	101	2026-04-01	500
R002	102	2026-04-10	1200
R003	103	2026-05-14	500
R004	104	2026-05-10	1500
R005	105	2026-06-28	2000

Return only orders that are more than 30 days old from today, calculate days_since_order using today's date

SELECT

transaction_id,

customer_id,

order_date,

amount,

datediff(current_date(), order_date) AS days_since_order

FROM

quarterly_transactions

WHERE

datediff(current_date(), order_date) > 30;

transaction	order date	days since order
Q001	2026/04/01	183
Q002	2026/04/10	174
Q003	2026/05/14	140
Q004	2026/05/10	113
Q005	2026/06/28	95